

ARYA ZADEBRAHIMI

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PERSONAL WEBPAGE

EDUCATION

Aug 2024

Master of Science in Bioinformatics

Sharif University of Technology

- Cumulative GPA of 3.9 (18.2 / 20)
- Relevant coursework: Synthetic Biology, Reinforcement Learning, Machine Learning for Bioinformatics

May 2021

Bachelor of Science in Software Engineering

Azad University

- Cumulative GPA of 3 (15.58 / 20)
- Relevant coursework: Probability and Statistics, Artificial Intelligence, Data Mining, Algorithm Design

RESEARCH INTEREST

- Microbial Communities
- Metabolic Engineering
- Graph Neural Networks
- Reinforcement Learning

RESEARCH EXPERIENCE

Constraint-based Optimization to Increase By-product Yield

- Modification of a previous work (SteadyCom)
- Ensures maximum yield and robustness by using microbial community
- Failed due to high computational costs and impracticality

Organism Flux Prediction using Graph Neural Network

- Trained a GNN model on microbes' metabolic networks in a community
- Achieved a Root Mean Square Error of 2 between predicted flux and Flux Balance Analysis results
- Currently reducing loss by adding meaningful features to each reaction

PROJECTS

- Analysis of microarray gene expression data to identify genes involved in AML (Acute Myeloid Leukemia)
- Docking and optimizing the conformation of the Fruquintinib inhibitor molecule
- A transformer-based model to predict financial market price
- Single-cell RNA sequencing clustering and analysis

TEACHING ASSISTANT

Generative Model Design

- Assignment Design
- Test Question Design

Biomedical Image Processing

- Assignment Design
- Test Question Design

FURTHER EXPERIENCE

Aug 2022

Wet Lab Techniques and Molecular Biology

Iran University of Medical Sciences, Tehran

- Short-term internship
- Cell DNA extraction
- DNA detection using PCR
- Cell culture
- Cell counting using microscope

Oct 2017 – Dec 2018

Android Developer

Tehran

- Developed 5 separate projects
- Scrum team collaboration framework

SKILLS

TOEFL: 104 (Reading 30, Listening 28, Speaking 23, Writing 23)

Programming: Python (pytorch, pytorch geometric, transformers, numpy, pandas, opencv), Julia, Java, R, Android, TCL, C#, React